

Homework 1

ASEN 5335 Aerospace Environment

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Problem 1: Radiation from Corona

(i)

Given T of the corona region to be $T = 2\text{MK}$. We can use the equation,

$$\lambda_{max}T = \alpha$$

Where $\alpha = \text{constant}(2.98 * 10^{-3}\text{K})$.

Rearrange to solve for wavelength.

$$\lambda_{max} = \frac{T}{\alpha}$$

The corona region's temperature can be substituted and solved.

$$\lambda_{max} = 1.449 * 10^{-9}\text{m} = 14.49 * 10^{-10}\text{m}$$

This would also be in the same range for X-rays, $\approx 10^{-10}\text{m}$.

(ii)

For the region photosphere region at a cooler $T = 5,778\text{K}$, the process is the same.

$$\lambda_{max} = 5.01 * 10^{-7}\text{m} = 50.15 * 10^{-8}\text{m}$$

This would be in the same range for UV $\approx 10^{-8}\text{m}$.

Problem 2: Thermal Escape

From Wikipedia we can obtain an equation for most probable thermal velocity of a particle.

$$v_{th} = \sqrt{\frac{2k_B T}{m}}$$

$$k_B = 1.38 \times 10^{-23}\text{J/K}$$

$$m = m_{\text{proton}} = 1.673 \times 10^{-27}\text{kg}$$

$$T = 2\text{MK} = 2 \times 10^6\text{K}$$

$$\therefore v_{th} = 181,644\text{m/s}$$

We then calculate the escape velocity using the equation,

$$v_{escape} = \sqrt{\frac{2GM}{r}}$$

$$G = 6.674 * 10^{-11} \text{ Nm}^2/\text{kg}^2$$

$$M = 1.989 * 10^{30} \text{ kg}$$

$$r = 10 * 10^6 * 10^3 \text{ m}$$

Where G is the universal gravitational constant, M is the mass of the sun, and r is the radius of the of the outer corona. We then find,

$$v_{escape} = 163,000 \text{ m/s}$$

$$\because v_{th} < v_{escape}$$

It is possible for the coronal protons to escape the sun's gravity

Problem 3: Solar Wind

(i.)

Given quiescent solar wind density $n = 5 \text{ ions/cm}^3$, and $V = 400 \text{ km/s}$ at Earth, we can use the mass flow equation

$$\dot{m} = \rho * V * A$$

To find A , we calculate the surface area of a sphere given the radius of Earth,

$$A = 4\pi r^2$$

$$r_{Earth} = 149.6 * 10^9 \text{ m}$$

$$\therefore A = 1.87 * 10^{12} \text{ m}^2$$

Convert the ion density to density in terms of kg/m^3 , given the mass of a proton being $m_{proton} = 1.67 * 10^{-27} \text{ kg/ion}$, we calculate,

$$\dot{m} = 5 \text{ ions/cm}^3 * 1.67 * 10^{-27} \text{ kg/ion} * \frac{1 \text{ cm}^3}{1 * 10^{-6} \text{ m}^3} * 400 * 10^3 \text{ m}$$

$$\dot{m} = 939,332,865.8 \text{ kg/s}$$

(ii.)

Finding the flux at the orbit of Jupiter, we are able to use the inverse square relation. We find flux at Earth,

$$\phi_{Earth} = \rho * V$$

$$\phi_{Earth} = \frac{5 \text{ ions}}{1 * 10^{-6} \text{ m}^3} * 400 * 10^3 \text{ m/s}$$

$$\phi_{Earth} = 2 * 10^{12} \text{ ions/m}^2/\text{s}$$

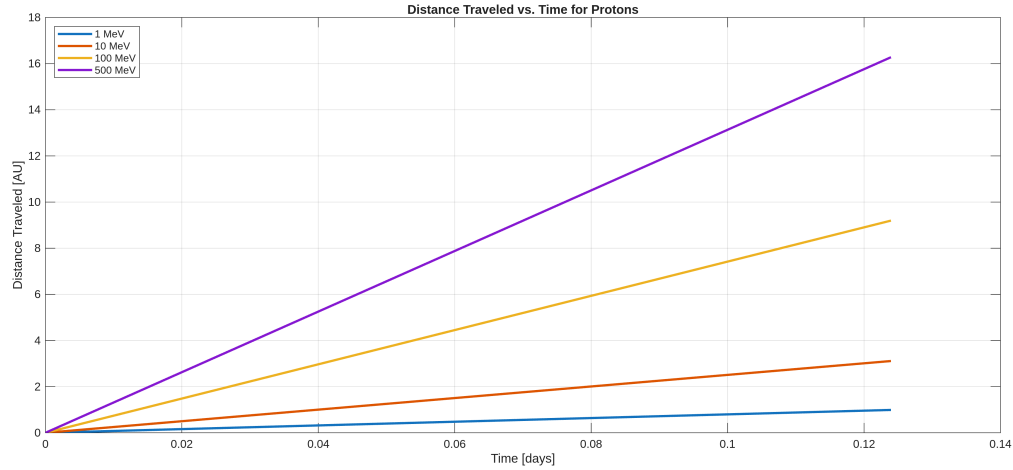


Figure 1: AU traveled vs time

With inverse square law if $r_{Earth} = 1\text{AU}$ and $r_{Jupiter} = 5.2\text{AU}$

$$\phi_{Jupiter} = \phi_{Earth} * \frac{1}{5.2^2}$$

$$\phi_{Jupiter} = 7.396 * 10^{10} \text{ ions/m}^2/\text{s}$$

$$\phi_{Jupiter} = 7.396 * 10^{14} \text{ ions/cm}^2/\text{s}$$

Problem 4: Relativistic Protons

Plotting distance traveled in AU w.r.t for multiple protons of varying MeV.

To solve for how long it takes in hours to for each of these energies to each Earth, we are able to use,

$$KE = \left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} - 1 \right) m_0 c^2$$

where we can convert MeV to Joules with,

$$MeV = 1.602 * 10^{-13} \text{ J}$$

This can be rearranged to,

$$V = \sqrt{1 - \frac{1}{\left(\frac{KE}{m_0 c^2} + 1\right)^2}}$$

We are able to use this equation to calculate all the velocities given the MeVs. Assuming this velocity is constant through the process, we can simply find some distance d , which is the distance

between the lower corona of the Sun and the Earth,

$$\begin{aligned}
 d &= 1\text{AU} - 2R_S \\
 1\text{ AU} &= 1.496 * 10^{11}\text{m} \\
 1 R_s &= 6.957 * 10^8\text{m} \\
 \therefore d &= 1.4821 * 10^{11}\text{ m}
 \end{aligned}$$

Which time t can be calculated with simply $t = d/v$.

	1 MeV	10 MeV	100 MeV	500 MeV
Velocity [m/s]	1.384e+07	4.345e+07	1.285e+08	2.274e+08
Time [s]	1.071e+04	3.411e+03	1.154e+03	6.518e+02
Time [hr]	2.975	0.947	0.320	0.181

Table 1: Properties of Lower Corona Proton at varying MeV

Problem 5: Heliopause

In order to calculate the heliopause we first seek the pressure as a result of the velocity and ion density at Earth. The inverser square law can then be applied to solve for the heliopause distance.

$$\begin{aligned}
 P_{Earth} &= n_{Earth}mv^2 \\
 n_{Earth} &= 5\text{ ions/cm}^3 = 0.005\text{ ions/m}^3 \\
 m &= 1.673 * 10^{-27}\text{ kg/ion} \\
 v &= 400 * 10^3\text{ m/s} \\
 \therefore P_{Earth} &= 1.3384 * 10^{-21}
 \end{aligned}$$

(Problem 6: Spacecraft Temperature)

(i.)

Given aluminum with $\alpha = 0.15$ and $\epsilon = 0.05$, to find equilibrium temperature at Earth, we use $r = 149.6 * 10^9\text{m}$ and then use the equation for thermal equilibrium,

$$\begin{aligned}
 T &= \left(\frac{\alpha}{\epsilon}\right)^{\frac{1}{4}} \left(\frac{SA_n}{\sigma A_{tot}}\right)^{\frac{1}{4}} \\
 S &= 1361\text{ W/m}^2 \\
 \sigma &= 5.67 * 10^{-8}\text{ W/m}^2/\text{K}^4 \\
 A_n &= \pi(149.6 * 10^9)^2\text{ m}^2 \\
 A_{tot} &= 4\pi(149.6 * 10^9)^2\text{ m}^2 \\
 \therefore T &= 366.3\text{ K}
 \end{aligned}$$

Where S is solar flux, σ is the Stefan-Boltzmann constant, A_n is the cross sectional area normal to the Sun, and A_{tot} is the surface area at Earth's orbit.

(ii.)

For Jupiter, we modify the r used to be the r at Jupiter. The process remains the same substituting $r = 778 * 10^9 \text{ m}$

$$\therefore T = 366.3 \text{ K}$$

Interestingly it remains very close to the thermal equilibrium for Earth as well.

(iii.)

When repeating the process for GSFC White Paint NS-74 with $\alpha = 0.17$ and $\epsilon = 0.92$, we get,

$$T = 182.5 \text{ K}$$

I think this paint was designed with the explicit purpose of being able to have a lower equilibrium temperature as a result of radiation from the Sun for satellite missions. High temperatures can be detrimental to equipment onboard and the strategy to paint the spacecraft with paint that improves its emissivity by a considerable amount would be very useful in the space environment by lowering the equilibrium temperature.

Problem 7: Solar Outputs